

E.1 Structure of the atom

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

$h = 6.63 \times 10^{-34} \text{ J s}$; $c = 3.0 \times 10^8 \text{ m s}^{-1}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. Show your working.

Question 1 — The neon sign writer

(9 marks)

A sign-maker's gas discharge tube glows with a spectrum of sharp, separate colours rather than a continuous rainbow.

- (a) Explain how discrete atomic energy levels produce a spectrum of discrete lines. [3]
- (b) An atom in the tube has exactly four energy levels. Calculate the maximum number of different spectral lines it can emit. [2]
- (c) The sign-maker wants the reddest possible line from these levels. State which transition produces the longest wavelength, in terms of the size of the energy gap. [2]
- (d) An apprentice counted “four levels, four lines”. State what should have been counted instead. [2]

Question 2 — Pricing a photon

(5 marks)

One transition in the tube releases a photon of energy 3.0 eV .

- (a) Calculate the photon's wavelength, in nanometres. [3]
- (b) A colleague reasons “more energy means longer wavelength, so this is a deep red line”. Correct the direction of the relationship and classify the actual colour region. [2]

Question 3 — The gold-leaf ambush

(7 marks)

In the Geiger–Marsden experiment, alpha particles were fired at a thin gold foil.

- (a) State the two key observations. [2]
- (b) State the conclusion drawn from each observation about the structure of the atom. [4]
- (c) Explain why a “plum pudding” atom could not have produced the rare large-angle deflections. [1]

Question 4 — Two spectra, one atom

(4 marks)

The same element can show a bright-line *emission* spectrum or a dark-line *absorption* spectrum.

- (a) Describe how each spectrum is produced. [3]
- (b) State the relationship between the line positions of the two spectra for the same element. [1]

Answers

Question 1 — The neon sign writer

- (a) An electron in an atom may occupy only certain discrete energy levels. When it drops from a higher level to a lower one, the energy lost is carried off by a single photon whose energy exactly equals the gap, $E = hf$. Because the gaps are discrete, only certain photon energies are possible, so only certain frequencies (and hence colours) appear: sharp separate lines rather than a continuous rainbow.
- (b) A line corresponds to a *pair* of levels, and any of the four may be the upper with any lower one: $\frac{4 \times 3}{2} = 6$ lines.
- (c) The transition with the *smallest* energy gap. Since $E = hc/\lambda$, energy and wavelength are inversely related, so the least energetic transition gives the longest wavelength, the reddest line.
- (d) The number of *pairs* of levels, that is, the number of possible transitions between them, rather than the number of levels. Four levels give six pairs, not four.

Question 2 — Pricing a photon

- (a) $E = 3.0 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-19}$ J, so $\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{4.8 \times 10^{-19}} = 4.1 \times 10^{-7}$ m = 410 nm.
- (b) The relationship is *inverse*, not direct: $\lambda = hc/E$, so more energy means a *shorter* wavelength. At 410 nm this line sits at the violet end of the visible spectrum, about as far from deep red as visible light goes.

Question 3 — The gold-leaf ambush

- (a) (1) The great majority of alpha particles passed straight through the foil with little or no deflection. (2) A very small fraction, of order one in eight thousand, were deflected through very large angles, some almost straight back.
- (b) (1) The atom is mostly empty space, since most alphas pass through undisturbed. (2) There is a tiny, dense, positively charged nucleus carrying almost all the atom's mass, since only a concentrated charge could repel a fast alpha through such a large angle.
- (c) Spreading the positive charge thinly through the whole atom gives only weak electric fields everywhere, so no single encounter could deliver the large impulse needed to reverse a fast alpha particle. The rare large-angle deflections therefore rule the model out.

Question 4 — Two spectra, one atom

- (a) *Emission*: atoms excited by heating or by a discharge drop back down between levels and emit photons at their transition energies, giving bright lines on a dark background. *Absorption*: white light with a continuous spectrum passes through the cooler gas, which absorbs exactly those same energies to excite its electrons, then re-radiates them in all directions rather than straight on. The result is dark lines on a continuous background.
- (b) The lines lie at exactly the *same* wavelengths, because both are set by the same set of energy level gaps in that element. This is what makes spectral lines a fingerprint for identifying elements.